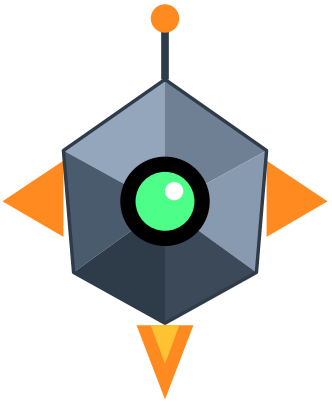
BOOK 6

What Must Remain Unresolved

Saying what you do know





MULTIX//HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

Pop What You Can

STORY

Tavi had a jar of marbles on the table. The lid was stuck, and nobody knew how many marbles were inside. Next to the jar sat 3 small bags, and each bag held 4 marbles. Tavi's little sister asked, "How many marbles do you have altogether?"
"I don't know yet," said Tavi, "because I can't open the jar. But I can already do part of the work."

STEP 1 Draw circles

Let's call the number of marbles in the jar **x**. It is a mystery number.

Tavi wrote down the whole total, and then drew a circle around each piece that belongs together. In this book, a pair of round brackets is a circle.

TRACE

```
( x + ( 3 × 4 ) )
```

There are two circles here, one inside the other. The inner circle holds the bags: 3 bags of 4. The outer circle holds the whole total: the jar plus the bags.

STEP 2 The rules of popping

Tavi has three rules.

1. **A circle is ready when everything needed for its calculation is known and its operation is allowed.** A mystery number or an unfinished smaller circle inside means it is not ready.
2. **Popping a circle replaces it with its value.** Everything outside the circle stays exactly as it was.
3. **If no circle is ready, you stop.** What is left is your note to yourself.

Trust check. A pop is trustworthy because rule 2 is exact: the circle (3 × 4) really does have the value 12, and nothing outside it was touched.

STEP 3 Pop the circle that is ready

Which circle is ready? The inner one. It holds only the plain numbers 3 and 4. The outer one holds x, so it must wait.

Tavi pops the inner circle. Here is the notebook, with a line for each pop:

TRACE

```
( x + ( 3 × 4 ) )
pop ( 3 × 4 )  →  12
( x + 12 )
```

Now Tavi checks rule 3. The only circle left holds x, so it is not ready. Tavi is **stuck**, and that is allowed. Being stuck does not mean something went wrong. It means every bit of work that did not need the jar has been done.

What is left, (x + 12), is the note to self: *"Whatever is in the jar, add 12."*

When the jar finally opens and there are 20 marbles inside, Tavi puts 20 in place of x. Now 20 + 12 is ready to pop, and the answer is 32.

Test the note: Does it match the original? With x = 20, the original is 20 + 3 × 4 = 32, and the note gives 20 + 12 = 32. They agree.

STEP 4 What a pop forgets

When Tavi popped the circle (3 × 4), something was lost. The 12 does not remember that it came from 3 bags of 4. It could have been 2 bags of 6, or 12 bags of 1.

This is the same thing that happened to Lena's calculator in Reversing the Calculation: once the answer is on the screen, the road that led to it is gone.

The notebook solves this. Each line of the trace says *what was popped* and *what it became*. The circles forget, but the notebook remembers.

STEP 5 Which circle first?

Sometimes more than one circle is ready. Look at this one:

TRACE

```
( ( 2 + 3 ) × ( 4 + 1 ) )
```

Both inner circles hold only plain numbers, so either can be popped first.

Pop the left one first:

TRACE

```
( ( 2 + 3 ) × ( 4 + 1 ) )
pop ( 2 + 3 )  →  5
( 5 × ( 4 + 1 ) )
pop ( 4 + 1 )  →  5
( 5 × 5 )
pop ( 5 × 5 )  →  25
```

Pop the right one first:

TRACE

```
( ( 2 + 3 ) × ( 4 + 1 ) )
pop ( 4 + 1 )  →  5
( ( 2 + 3 ) × 5 )
pop ( 2 + 3 )  →  5
( 5 × 5 )
pop ( 5 × 5 )  →  25
```

Both trails end at 25. For this problem the order did not matter, though the two notebooks look different. Sometimes the order changes how many steps you take, and later chapters will look at that.

STEP 6 What popping cannot do

Now try this one:

TRACE

```
( ( x + 3 ) × 4 )
```

Is any circle ready? The inner circle holds x. The outer circle holds a circle, so it is not ready either. Tavi is stuck at the very start.

Popping alone cannot make progress here. That is not a flaw in the circles. It means this problem needs a **redraw** first, such as sharing the "× 4" with everything inside the inner circle:

TRACE

```
( ( x + 3 ) × 4 )
redraw  →  ( ( x × 4 ) + ( 3 × 4 ) )
pop ( 3 × 4 )  →  12
( ( x × 4 ) + 12 )
```

The redraw keeps the value, because four lots of (x + 3) is four lots of x and four lots of 3. After it, one circle is ready, and Tavi can pop it. The circle with x in it must still wait.

Notice also that (x + (3 × 4)) and ((x + 3) × 4) are different problems, because the circles are in different places. With x = 5, the first gives 5 + 12 = 17 and the second gives 8 × 4

= 32. The circles tell you what belongs together, so moving them changes the question.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the popping keep?** (The total, for every possible x. Test it with a number.)
- **What did a pop forget?** (How the value was made. The 12 forgot it was 3 bags of 4.)
- **What did Tavi do ahead of time, and what was left?** (Tavi popped the circle with plain numbers and left the one holding x.)

WHAT MAKES THIS TRUSTWORTHY?

A pop is trustworthy because rule 2 is exact: the circle (3 × 4) really does have the value 12, and nothing outside it was touched. So the note (x + 12) must agree with the original for every x, and that is a reason, not just a test.
Testing with a number, as we did with x = 20, is a second, separate kind of evidence. It can catch a slip in the notebook.

PRACTICE

For each one, write out the notebook with a line for each pop. If you get stuck, say what is left.

1. `( x + ( 2 × 6 ) )`
2. `( ( 5 + 4 ) × x )`
3. `( ( 10 − 4 ) + x )`
4. `( ( 3 + 2 ) × ( x + 1 ) )`
5. `( ( 6 − 1 ) × ( 3 + 4 ) )`. Pop it in two different orders. Do you get the same answer?
6. Dani says that (x + (3 × 4)) is the same as ((x + 3) × 4). Use x = 5 to show that Dani is wrong. Which circle is in the wrong place?

BRIDGE *Bridge to the rest of the series:* In Reversing the Calculation, Lena could not tell which road led to 12, and a pop forgets in the same way. The notebook of pops is the "diary" from that chapter, written at the level of one circle per line.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  6
6 pop ( 1,600 − 6 )  →  1,594
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × 4 ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × 4 )  →  16
6 pop ( 2,500 − 16 )  →  2,484
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 2

Two Jars

STORY

Tavi's sister found a second jar. Now there were two sealed jars on the table. Jar x was the first one, and the new one was jar y. Next to jar x were 3 bags of 4 marbles. Next to jar y were 2 bags of 5 marbles.
"How many marbles are there altogether?" asked the sister.
"I can't say yet," said Tavi. "But I can do a lot of it."

STEP 1 Pop what you can

Tavi drew a circle around each jar with its bags, and one big circle around everything:

TRACE

```
( ( x + ( 3 × 4 ) ) + ( y + ( 2 × 5 ) ) )
```

Which circles are ready? A circle is ready when everything it needs is known. Two inner circles are ready: (3 × 4) and (2 × 5). The circles with x and y are blocked.

TRACE

```
( ( x + ( 3 × 4 ) ) + ( y + ( 2 × 5 ) ) )
pop ( 3 × 4 )  →  12
pop ( 2 × 5 )  →  10
( ( x + 12 ) + ( y + 10 ) )
```

Now Tavi is stuck again. But this time the stuck pile is bigger, and it has two mysteries.

STEP 2 Redraw so the plain numbers meet

The 12 and the 10 are plain numbers, but they are in different circles. They cannot be popped together. Tavi can redraw, because adding in a different order gives the same total:

TRACE

```
( ( x + 12 ) + ( y + 10 ) )
redraw  →  ( ( x + y ) + ( 12 + 10 ) )
pop ( 12 + 10 )  →  22
( ( x + y ) + 22 )
```

Trust check. The notebook is trustworthy because every pop is of plain numbers, and every redraw keeps the value for any x and y.

This is the shortest note Tavi can write: *“The two jars together, plus 22.”* The mysteries x and y stay inside one circle. Nothing in that circle can be popped without knowing the jars.

Tavi can tell the sister something real: “There are at least 22 marbles, and the jars add some more.”

STEP 3 One clue helps, a different one doesn’t

The sister reads the labels: “These two jars together hold 15 marbles.”

That is a clue about the circle (x + y). It does not say what x is or what y is, but it says what they add up to, which is all that circle needs.

TRACE

```
( ( x + y ) + 22 )
use the clue ( x + y ) = 15
( 15 + 22 )
pop  →  37
```

The total is **37**, even though Tavi still does not know x or y separately. They could be 7 and 8, or 10 and 5, or 15 and 0. The answer does not change.

Now suppose the label had said only “jar x holds 6 marbles.” That is a clue about x, but not about y, so (x + y) is still blocked. The total would be ((6 + y) + 22), which is (y + 28), and Tavi would still be stuck.

The lesson: a clue helps only if it matches the shape of the blocked circle.

STEP 4 Some mysteries vanish

Look at this circle:

TRACE

```
( x × 0 )
```

It holds a mystery x, so it does not look ready. But Tavi knows that anything times 0 is 0, whatever x is. So this circle can be redrawn into a plain **0**, and the mystery disappears.

TRACE

```
( x × 0 )
redraw  →  0
```

Not every mystery can vanish this way. (x + 0) simply becomes x, and (x × 1) simply becomes x. But (x × 0) is special.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the redraw in Step 2 keep?** (The total, for any x and any y.)
- **What did the clue (x + y) = 15 keep, and what did it leave unknown?** (It kept the sum of the jars. It left x and y unknown.)

WHAT MAKES THIS TRUSTWORTHY?

The notebook is trustworthy because every pop is of plain numbers, and every redraw keeps the value for any x and y. So the note ((x + y) + 22) is right for every pair of jars. You can test it with a pair, like x = 7 and y = 8: the original is (7 + 12) + (8 + 10) = 37, and the note is 15 + 22 = 37. ✓
A test with one pair of numbers is evidence. The reason is that each redraw kept the value.

PRACTICE

1. Pop and redraw ((x + (2 × 3)) + (y + (4 × 2))) as far as you can. What is the note to self?
2. If x + y = 10, what is the total in problem 1?
3. Is this clue enough to finish problem 1: “x is 4”? Why or why not?
4. Simplify ((x × 0) + (y + 5)).
5. Dani says the clue “x + y = 15” lets us know that x is 7. Is Dani right?
6. A jar x holds a number, and Tavi writes ((x + 5) − 5). Redraw it to something simpler. Why does it work?

BRIDGE *Bridge to the rest of the series:* In chapter 1, one mystery gave one stuck circle. Here two mysteries share a circle, and a clue about their sum finishes the job. In Reversing the Calculation, Omar’s ×0 button cannot be undone for the same reason that (x × 0) forgets x.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **3,599**. Leave the others blank. What appears?

3584 + 20	3573 + 21	3581 + 23	3630 − 31	3601 − 2	3587 + 11	38 × 95	3594 + 6
3587 + 6	3583 + 22	93 × 93 − 71 × 71	3573 + 26	3632 − 33	3592 + 9	2 × 1804	3582 + 19
3586 + 24	61 × 59	3619 − 20	3587 + 12	3603 − 4	3590 + 9	3627 − 28	3570 + 17
3576 + 23	3577 + 22	3589 + 13	3567 + 32	3625 − 26	3577 + 14	3581 + 18	3614 − 15
3571 + 28	3564 + 35	3579 + 20	3578 + 21	3632 − 33	3608 − 9	3613 − 14	3572 + 27
3584 + 15	3564 + 35	3568 + 31	3611 − 12	3583 + 16	3568 + 31	3626 − 27	3607 − 8
3576 + 13	3609 − 10	3617 − 18	3585 + 21	3574 + 16	3571 + 28	3638 − 39	48 × 75
3579 + 11	3591 + 8	3561 + 38	65 × 65 − 25 × 25	3586 + 22	3579 + 20	3575 + 24	46 × 78



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 3

Notes That Tell the Truth

STORY

When Tavi gets stuck, the notebook ends with a note to self. That note matters, because someone else might use it. Tavi, Dani and Sam each wrote a note for the same circle, and the notes did not agree.
The circle was this:

TRACE

```
( ( x + 3 ) × 4 )
```

The jar x is sealed. Each wrote what they could.

STEP 1 Three notes

Tavi's note: ((x × 4) + 12)

Dani's note: ((x × 4) + 3)

Sam's note: (x + 12)

They cannot all be right. How can you tell which notes are faithful? A faithful note gives the same answer as the original **for every x**.

STEP 2 Test with numbers

You cannot test every x, but you can test a few. Drop a number in for x, then pop every circle.

x	Original ((x + 3) × 4)	Tavi	Dani	Sam
0	12	12	3	12
1	16	16	7	13
2	20	20	11	14

Look at each column.

- **Tavi** matches the original every time.
- **Dani** is wrong from the very first row. She shared the 4 with x but forgot to share it with the 3.
- **Sam** matches at x = 0 and then falls behind. If Tavi had tested only x = 0, Sam's note would have looked fine.

Trust check. A note is trustworthy when it has a reason (each redraw keeps the value) and when it passes tests at several different values.

STEP 3 What a test shows, and what it doesn't

One lesson: **x = 0 can hide mistakes.** Many wrong notes happen to agree when x is 0. A good test uses several numbers, including some that are not 0 or 1.

A second lesson: **a pass is only evidence**. Tavi's note passed three tests. What makes Tavi's note trustworthy is the reason:

TRACE

```
( ( x + 3 ) × 4 )
redraw  →  ( ( x × 4 ) + ( 3 × 4 ) )
pop  →  ( ( x × 4 ) + 12 )
```

The redraw shares the 4 with everything inside, so it keeps the value for every x. The tests are a safety net, but the reason is what holds the note up.

STEP 4 The worst note is a false pop

There is one more way to write a bad note. A friend tried to be helpful and popped too much, writing `( 7 + 12 )` → 19, as if x were 7. She did not know x. That is a pop that used something nobody knew.

A note to self must not claim more than the notebook knows. If x is not known, the note has to keep x. A shorter note is only better if it is still true.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does a faithful note keep?** (The value of the original, for every x.)
- **What can a careless note lose or invent?** (Part of the original, like a forgotten 3, or a number nobody knew.)

WHAT MAKES THIS TRUSTWORTHY?

A note is trustworthy when it has a reason (each redraw keeps the value) and when it passes tests at several different values. A note with only tests might hide a mistake at some value you did not try. A note with only a reason might hide a slip in the arithmetic. Use both.

PRACTICE

1. Test these notes for ((x + 2) × 5) at x = 0, 1 and 2: (a) `( ( x × 5 ) + 10 )`, (b) `( ( x × 5 ) + 2 )`, (c) `( x + 10 )`.
2. Which notes in problem 1 look fine at x = 0? Which of them fails at x = 1?
3. Write a faithful note for (2 × (x + 6)) and test it at x = 1.
4. Tavi writes ((x + 3) × 4) → (x + 12), saying “4 times 3 is 12”. What did Tavi forget?
5. A note says (x + 12) but a friend says x is probably 10, so writes 22. Is 22 a faithful note? Why or why not?

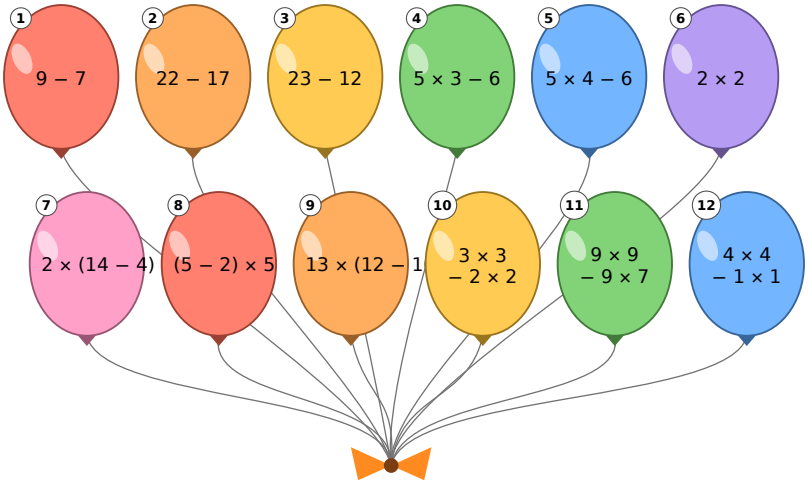
BRIDGE *Bridge to the rest of the series:* In chapter 2, the note ((x + y) + 22) was checked with one pair of jars. Here we see why one check is not enough: a wrong note can pass at x = 0. In Two Boxes, One Answer, the detective’s rule says the same about boxes: a finite test cannot prove two things are always equal.

PUZZLE PAGE

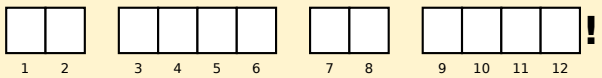
The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?



CHAPTER 4

Saying What You Do Know

STORY

Tavi still had not opened the jar. But her sister found a note on the lid: *between 5 and 9 marbles.*
Now Tavi knows something, but not everything. She wanted to say what she could without pretending.

STEP 1 A range for the jar

The jar x holds at least 5 and at most 9 marbles. The bags still hold 3 × 4 = 12.

TRACE

```
( x + ( 3 × 4 ) )
pop ( 3 × 4 )  →  12
( x + 12 )
```

Tavi is stuck on x, but she can still say something true about the total. Try the smallest and largest jar.

TRACE

```
smallest jar:  ( 5 + 12 )  →  17
largest jar:   ( 9 + 12 )  →  21
```

The total is **between 17 and 21**.

That is a good note. It does not claim to know the jar, and it does not throw away the clue. It is faithful, as in chapter 3, and more useful than “I don’t know.”

STEP 2 Different circles, different ranges

Tavi tried other circles with the same jar.

(x × 4): four lots of the jar. Smallest 5 × 4 = 20, largest 9 × 4 = 36. So the answer is between **20 and 36**.

(x + y): two jars, each between 5 and 9. Smallest 5 + 5 = 10, largest 9 + 9 = 18. So between **10 and 18**.

Trust check. A range is trustworthy when it is built from the clue and both ends: the smallest and largest jar.

STEP 3 Watch out for a flip

Then Tavi tried a circle where the jar is subtracted:

TRACE

```
( 30 − x )
```

The smallest jar gives the **largest** answer.

TRACE

```
smallest jar (5):  ( 30 − 5 )  →  25
largest jar (9):   ( 30 − 9 )  →  21
```

So (30 − x) is between **21 and 25**. If Tavi had just used the old order, 5 for the small end and 9 for the big end, she would have written “between 25 and 21”, which makes no sense.

The rule: **try both ends of the jar, and let the answers tell you which is small and which is large.**

STEP 4 A range is a kind of residual

In chapter 1, the note to self was (x + 12). Here the note has two parts: the expression (x + 12), and the clue about x. Together they say: “17 to 21 marbles, and I’ll know more when the jar opens.”

When the jar is opened and x turns out to be 7, Tavi pops (7 + 12) to 19. That is inside the range 17 to 21, so the range was honest.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the range keep?** (A true statement about every jar that fits the clue.)
- **What did it leave unknown?** (Exactly which jar it is.)

WHAT MAKES THIS TRUSTWORTHY?

A range is trustworthy when it is built from the clue and both ends: the smallest and largest jar. It is tested when the jar opens. A value outside the range means a mistake in the clue or the arithmetic.

PRACTICE

1. The jar x holds between 3 and 8. What is the range of (x + 12)?
2. The jar x holds between 5 and 9. What is the range

of (x × 3)?

3. The jar x holds between 5 and 9. What is the range of (50 − x)?

4. The jar x holds between 5 and 9, and the jar y holds between 2 and 4. What is the range of (x + y)?

5. The jar opens and x is 12. The note said “between 5 and 9”. What does that tell you?

BRIDGE *Bridge to the rest of the series:* In A Useful Wrong Answer, a bound gave a range that was certainly right. Here a clue about an unknown gives a range for a circle that cannot be popped yet. In Reversing the Calculation, a clue shrinks a pile in the same way.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  12
6 pop ( 1,600 − 12 )  →  1,588
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  36
6 pop ( 4,900 − 36 )  →  4,864
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 5

When the Total Is Known

STORY

Tavi's sister peeked at the receipt. "There are 20 marbles altogether," she said.
Now Tavi had an unknown jar and a known total. The circle was stuck at (x + 12), and she also knew it equals 20.

TRACE

```
( x + 12 )  =  20
```

STEP 1 Undo to find the jar

Tavi did not need to guess. She undid the +12, as in Reversing the Calculation.

TRACE

```
20 − 12  →  8
```

So x = **8**. Check by putting 8 back in the circle: (8 + 12) pops to 20. ✓

What changed? The circle was stuck because it held x. A known total gave Tavi a way to work backward through the circle until x stood alone.

STEP 2 More than one step

A harder one. Tavi found 4 identical sealed jars, each holding x marbles, plus 3 bags of 4 marbles. The total is 40.

TRACE

```
( ( x × 4 ) + 12 )  =  40
```

The last thing done to x was ×4, then +12. Tavi undid them in the opposite order, like socks and shoes.

- Undo the +12: 40 − 12 = 28.
- Undo the ×4: 28 ÷ 4 = 7.

So x = **7**. Check: ((7 × 4) + 12) → (28 + 12) → 40. ✓

Trust check. The check is putting x back into the original circle and popping.

STEP 3 When the circles are arranged differently

Look at this arrangement, which has the same pieces in different circles:

TRACE

```
( ( x + 3 ) × 4 )  =  40
```

The last thing done is the ×4, so undo that first.

- Undo ×4: 40 ÷ 4 = 10.
- Undo +3: 10 − 3 = 7.

So x = **7** again. Check: ((7 + 3) × 4) → (10 × 4) → 40. ✓

Two different circles gave the same x, but only because the totals happened to line up. The circles are not the same question, as Tavi learned in chapter 1. They were set up to give the same answer, not to be the same.

STEP 4 Not always enough

Suppose the receipt said only "x + y = 15" for two jars. Tavi cannot find x and y from that. The total is known, but there are two unknowns in one circle, and many pairs fit.

A known total opens a circle only if there is **one** unknown inside it. With two, you need another clue.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the known total give Tavi?** (A way to work backward until x stood alone.)
- **What did she still need to check?** (That the x she found goes back into the original circle and gives the total.)

WHAT MAKES THIS TRUSTWORTHY?

The check is putting x back into the original circle and popping. If it gives the known total, the answer fits. That check does not tell you whether another x could also fit. Here it could not, because each undo step has only one answer.

PRACTICE

1. Find x in (x + 7) = 30.
2. Find x in ((x × 3) + 5) = 29.
3. Find x in ((x − 4) × 2) = 18.
4. Check your answer to problem 3 by putting it back.
5. The receipt says (x + y) = 20. Can Tavi find x? What would help?

BRIDGE *Bridge to the rest of the series:* In chapter 4, a clue gave a range for an unknown. Here a known total gives one exact value, using the undo from Reversing the Calculation. In Solving by Changing the Question, finding where a gap is zero is another way to reach the same x.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,475**. Leave the others blank. What appears?

2449 + 27	2470 + 11	2462 + 24	2503 − 28	2498 − 23	2444 + 19	2438 + 29	2459 + 22
2453 + 26	2464 + 16	2472 + 7	2451 + 24	2511 − 36	2462 + 25	2441 + 24	2438 + 26
2464 + 23	2440 + 35	2438 + 37	2509 − 34	2491 − 16	2495 − 20	2455 + 20	2457 + 21
2505 − 30	2487 − 12	2465 + 12	9 × 275	2471 + 4	2469 + 15	2458 + 17	9 × 275
2504 − 29	2450 + 25	2448 + 27	54 × 54 − 21 × 21	2438 + 37	2466 + 9	2487 − 12	2460 + 15
2456 + 19	165 × 15	2456 + 19	2496 − 21	2459 + 16	2512 − 37	2452 + 23	2508 − 33
2466 + 14	45 × 55	2464 + 11	2461 + 21	2438 + 27	2449 + 26	99 × 25	2458 + 11
2451 + 17	5 × 495	2453 + 22	2475 + 2	2462 + 9	2494 − 19	9 × 275	2461 + 18



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 6

Two Notes That Are the Same

STORY

Tavi's friend Sam also had a stuck circle with a sealed jar, and he wrote his note differently from Tavi's.
Tavi's note: `( ( x × 4 ) + 12 )`
Sam's note: `( 4 × ( x + 3 ) )`
"Mine is simpler," said Sam. "Yours is simpler," said Tavi. Which note is right?

STEP 1 Test both with numbers

They tried a few jars.

x	Tavi's note	Sam's note
0	12	12
1	16	16
2	20	20
5	32	32

The notes agree every time. Are they the same?

STEP 2 Find the reason

Numbers are evidence. For a reason, Tavi redrew Sam's note:

Trust check. Two notes are the same when one can be redrawn into the other, and every redraw keeps the value.

TRACE

```
( 4 × ( x + 3 ) )
redraw  →  ( ( 4 × x ) + ( 4 × 3 ) )
pop ( 4 × 3 )  →  12
( ( 4 × x ) + 12 )
```

Four lots of x is the same as x taken four times, so (4 × x) and (x × 4) are the same. That gives Tavi's note. Because every redraw keeps the value, the two notes are the same for **every** jar.

They are the same note in different clothes.

STEP 3 Different clothes, different uses

If the notes mean the same, why would anyone prefer one?

- Sam's note (4 × (x + 3)) shows the **structure**: four lots of "the jar plus 3". It is shorter, and it tells you quickly that the answer is a multiple of 4.
- Tavi's note ((x × 4) + 12) shows the **pieces**: the part that depends on the jar, and the part that does not. It is easier to read off what the jar contributes.

The choice depends on what you need next. If Tavi is going to find x from a known total, as in chapter 5, either works. If she wants to say "the answer is 12 more than four jars", hers is clearer.

STEP 4 Two notes that only look the same

Compare these notes for two other circles.

Note A: (x + x) **Note B:** (x × x)

At x = 2, both give 4. But at x = 3, A gives 6 and B gives 9.

x	(x + x)	(x × x)
0	0	0
1	2	1
2	4	4
3	6	9

They agree at 0 and 2, and nowhere else. These notes are different. A test with only x = 2 would have missed it, which is the same lesson as the Two Boxes chapter.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What do two equal notes keep?** (The same value for every x.)
- **What can differ?** (The shape, which tells you different things.)

WHAT MAKES THIS TRUSTWORTHY?

Two notes are the same when one can be redrawn into the other, and every redraw keeps the value. Testing a few numbers can show they are different, but only the redraw shows they are the same for all jars.

PRACTICE

1. Redraw (2 × (x + 5)) into the shape ((2 × x) + something).
2. Is ((2 × x) + 5) the same note as (2 × (x + 5))? Test x = 0.
3. Is (x + x) the same note as (2 × x)? Give a reason.
4. Tavi's note is ((x × 4) + 12). Write another note for it that shows the structure "four lots of something".
5. For which values of x do (x + x) and (x × x) agree?

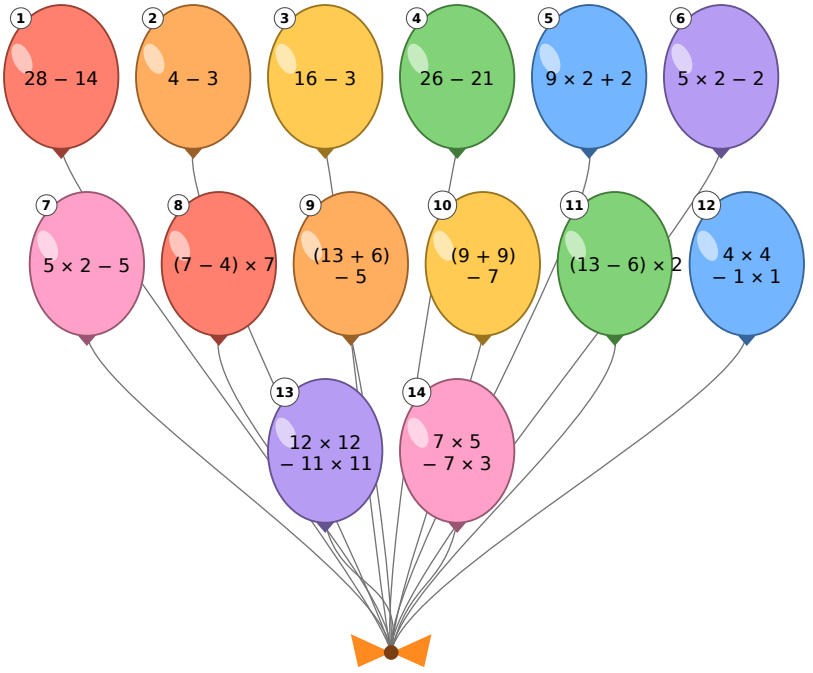
BRIDGE *Bridge to the rest of the series:* In chapter 3, a note had to be faithful to the original. Here two faithful notes are compared with each other. In Two Boxes, One Answer, two boxes that agree on a few cards can still differ, and a redraw is what shows two things really are the same.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 7

A Fraction Is a Note to Self

STORY

Tavi shared 17 marbles among 5 friends. Each friend got 3 marbles, and 2 were left over.
"What if I want to share the last 2 as well?" asked Tavi.

STEP 1 The circle that does not finish in whole numbers

Tavi wrote the problem in a circle:

TRACE

```
( 17 ÷ 5 )
```

In whole numbers, the answer is 3 with 2 left over. She wrote it as

TRACE

```
( 3 + ( 2 ÷ 5 ) )
```

The leftover circle (2 ÷ 5) holds only plain numbers, so it is ready. But 2 does not divide by 5 into a whole number. Tavi can leave it as a **fraction**, written 2/5, which means "2 shared among 5".

A fraction is a note to self. It says: *what is left is 2 shared among 5.* Nothing was lost, and nothing was pretended.

STEP 2 Popping to a decimal

Tavi could also pop the circle if decimals are allowed.

TRACE

```
( 2 ÷ 5 )  →  0.4
```

So 17 ÷ 5 = 3.4. Check: 5 × 3.4 = 17. ✓

Here the decimal is exact, so the pop loses nothing.

Trust check. A fraction is trustworthy because it says exactly what was left, and it can be popped later if a decimal is wanted.

STEP 3 When popping cannot finish

Now share 10 among 3 friends.

TRACE

```
( 10 ÷ 3 )  →  ( 3 + ( 1 ÷ 3 ) )
```

Each friend gets 3, and 1 is left over. Try popping the leftover:

TRACE

```
( 1 ÷ 3 )  →  0.3333...
```

The threes never stop. A pop cannot finish, because no ending decimal equals one third exactly. If Tavi writes 0.333 and multiplies by 3, she gets 0.999, not 1.

TRACE

```
( 3 × 0.333 )  →  0.999
( 3 × ( 1 ÷ 3 ) )  →  1
```

The fraction 1/3 keeps the answer exact. The decimal is only an approximation. So the faithful note is the fraction, and Tavi should keep it unpopped when she needs an exact answer.

STEP 4 Exact or approximate?

This is a choice, like the ones in The Boundary of a Step.

- For a rough answer, a decimal like 3.33 is fine.
- For an exact answer, keep 3 and 1/3.

And when Tavi writes a decimal, she should say it is approximate, as in chapter 1 of A Useful Wrong Answer.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What does the fraction keep?** (The exact leftover, 1 shared among 3.)
- **What does the decimal 0.333 lose?** (A tiny piece, so 3 × 0.333 is not 1.)

WHAT MAKES THIS TRUSTWORTHY?

A fraction is trustworthy because it says exactly what was left, and it can be popped later if a decimal is wanted. The check is to multiply back: 3 × (3 and 1/3) is 9 and 1, which is 10. ✓

PRACTICE

1. Write 11 ÷ 4 as a whole number plus a fraction. Pop the fraction to a decimal.
2. Write 10 ÷ 3 as a whole number plus a fraction. Can the fraction be popped exactly?
3. Check 3 and 1/3, times 3, equals 10.
4. Write 22 ÷ 7 as a whole number plus a fraction. Check by multiplying back.
5. When is it fine to write 3.33 for 10 ÷ 3?

BRIDGE *Bridge to the rest of the series:* In chapter 1, a stuck circle with x was a note to self. Here a fraction is a note to self too: a circle that could be popped, but only by losing something. In Solving by Changing the Question, a leftover scaled with the divisor, and a fraction is the cleanest way to write it.
Next in this book: Chapter 8: Holding 2,500 Until the Correction Arrives.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × 2 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × 2 )  →  6
6 pop ( 1,600 − 6 )  →  1,594
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  1
6 pop ( 625 − 1 )  →  624
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × ( 4 + 1 ) ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × ( 4 + 1 ) )  →  20
6 pop ( 2,500 − 20 )  →  2,480
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 8

Holding 2,500 Until the Correction Arrives

STORY

Tomas's friend Aiko gave him a puzzle. "I am thinking of two numbers that multiply to something. Their middle is 50. Each is the same distance from 50, one below and one above. I will not tell you how far, but it is somewhere from 1 to 5."
Tomas knew the case file. If the distance is d, the product is a square minus a corner.

STEP 1 The honest note

Tomas started the trace and got stuck.

TRACE

```
( product )
redraw  →  ( ( 50 × 50 ) − ( d × d ) )
pop ( 50 × 50 )  →  2,500
( 2,500 − ( d × d ) )
```

The circle (d × d) is **not ready**, because d is an unknown. Tomas cannot pop it. That is allowed. He keeps the circle as his note to self:

TRACE

```
( 2,500 − ( d × d ) )
```

This is the most he can say, and it is true. A guess like "2,496" would not be a note. It would be a bet.

STEP 2 What can be said while the note is open?

Tomas can still say useful things. He lists what d could be.

Trust check. An open note is honest.

d	corner (d × d)	product
1	1	2,499
2	4	2,496
3	9	2,491
4	16	2,484
5	25	2,475

So the product is somewhere from **2,475 to 2,499**. He also knows it is **less than 2,500**, because the corner is always more than nothing.

STEP 3 Label each claim

Tomas labels the claims he could make.

- "The product is less than 2,500." **Has a reason.** The corner is at least 1, so something is always taken away.
- "The product is between 2,475 and 2,499." **Has a reason.** It comes from d being 1 to 5.
- "The product is 2,500." **Refuted.** It would need d to be 0, and Aiko said d is at least 1.
- "The product is 2,496." **Only seen so far.** It is true if d is 2, as in 48 × 52, but nothing says d is 2.

STEP 4 The correction arrives

Aiko reveals: "The distance is 2." Now the circle is ready.

TRACE

```
( 2,500 − ( 2 × 2 ) )
pop ( 2 × 2 )  →  4
pop ( 2,500 − 4 )  →  2,496
```

The note is paid, the number is **2,496**, and it is the same as 48 × 52. The trace shows exactly when the stuck state ended, and why.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did Tomas's open note keep?** (Everything true about the problem: the middle, and the shape of the answer.)
- **What did it leave out?** (The one thing nobody had told him, the distance.)

WHAT MAKES THIS TRUSTWORTHY?

An open note is honest. It claims only what the clues support, and every claim has a label. When the missing piece arrived, Tomas did not have to start again. He finished the same trace.

PRACTICE

1. The middle is 50 and d is from 1 to 3. What is the range for the product?
2. The middle is 50, and d is even and at most 5. What are the possible products?
3. Label each claim with *has a reason*, *only seen so far*, or *refuted*, when d is unknown but from 1 to 5: (a) “The product is more than 2,400.” (b) “The product is 2,491.” (c) “The product is more than 2,500.”
4. The middle is 30 and d is from 1 to 2. Write the note and find the range.
5. Why is writing “(2,500 − (d × d))” better than writing “about 2,490”?

BRIDGE *Bridge to the rest of the series:* In A Useful Wrong Answer, the debt on an estimate was known. Here it is not yet known, and the notebook keeps the debt open. In Reversing the Calculation, a clue about the distance is what finds the pair.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,496**. Leave the others blank. What appears?

2467 + 26	2498 + 7	2472 + 27	48 × 52	2513 − 17	2501 + 3	2493 + 14	2484 + 17
2489 + 16	2480 + 21	2498 + 7	2471 + 25	2489 + 7	100 × 25	2495 + 10	2486 + 20
2481 + 22	2463 + 33	2494 + 2	2502 − 6	2507 − 11	2532 − 36	2532 − 36	2493 + 10
2503 − 7	2530 − 34	2493 + 6	2468 + 28	6 × 416	2484 + 13	2481 + 15	8 × 312
2469 + 27	2501 − 5	2501 − 5	16 × 156	2509 − 13	2500 − 4	2521 − 25	2480 + 16
2499 − 3	2506 − 10	3 × 832	2505 − 9	2518 − 22	2465 + 31	2529 − 33	2490 + 6
2469 + 24	96 × 26	2458 + 38	2489 + 18	2495 + 3	78 × 32	24 × 104	2487 + 7
2481 + 6	2487 + 9	2460 + 36	2488 + 14	2478 + 17	110 × 110 − 98 × 98	2484 + 12	2481 + 18



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

Answers (turn the book around)

Chapter 1

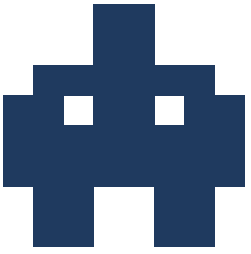
1. Pop (2 × 6) to get 12. Left: **(x + 12)**. Stuck, because of x.
2. Pop (5 + 4) to get 9. Left: **(9 × x)**. Stuck. Check with x = 2: the original gives 9 × 2 = 18, and the note gives 18.
3. Pop (10 − 4) to get 6. Left: **(6 + x)**. Stuck.
4. Pop (3 + 2) to get 5. Left: **(5 × (x + 1))**. The circle (x + 1) holds x, so it is not ready, and neither is the outer one. Stuck.
5. Pop (6 − 1) to get 5, then (3 + 4) to get 7, then (5 × 7) to get **35**. Pop (3 + 4) first, then (6 − 1), then (5 × 7): again **35**.
6. With x = 5, the first is 5 + 12 = 17 and the second is 8 × 4 = 32. They are different, so Dani is wrong. The circle around "x + 3" is in the wrong place: it takes the 3 away from the bags and gives it to the jar.

Spot the Slip.
Notebook A: slip on line **5**. 2 × 2 is 4. The answer should be 1,596.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 624.
Notebook C: **perfect**. 46 × 54 = 2,484.

Chapter 2

1. Pop (2 × 3) to 6 and (4 × 2) to 8. Redraw ((x + 6) + (y + 8)) to ((x + y) + (6 + 8)), pop to 14. Note: **((x + y) + 14)**.
2. (10 + 14) pops to **24**.
3. No. It tells about x only. The circle (x + y) still has y in it, so it cannot be popped.
4. Redraw (x × 0) to 0. Then (0 + (y + 5)) is (y + 5). The note is **(y + 5)**.
5. No. x could be 7 and y 8, but x could also be 10 and y 5. The clue fits many pairs.
6. It redraws to **x**. Adding 5 and then taking 5 away gives back what you started with, whatever x is.

Color by Answer (3,599). The shaded boxes make a robot.



Answers (turn the book around)

Chapter 3

1. The original gives 10, 15 and 20. (a) gives 10, 15, 20, so it passes. (b) gives 2, 7, 12, so it fails. (c) gives 10, 11, 12.
2. (a) and (c) both match at x = 0, with 10. (c) fails at x = 1: it gives 11, not 15.
3. `( ( 2 × x ) + 12 )`. At x = 1, the original is 2 × 7 = 14 and the note is 2 + 12 = 14. ✓
4. Tavi multiplied only the 3 by 4 and forgot to multiply the x. The correct note is ((x × 4) + 12).
5. No. 22 only fits x = 10, and nobody knows that. The note must keep x unless x is really known.

Balloon Message.

1.	9 − 7	= 2	B
2.	22 − 17	= 5	E
3.	23 − 12	= 11	K
4.	5 × 3 − 6	= 9	I
5.	5 × 4 − 6	= 14	N
6.	2 × 2	= 4	D
7.	2 × (14 − 4)	= 20	T
8.	(5 − 2) × 5	= 15	O
9.	13 × (12 − 10)	= 26	Z
10.	3 × 3 − 2 × 2	= 5	E
11.	9 × 9 − 9 × 7	= 18	R
12.	4 × 4 − 1 × 1	= 15	O

Message: **BE KIND TO ZERO!**

Chapter 4

1. 3 + 12 = 15 and 8 + 12 = 20, so **15 to 20**.
2. 5 × 3 = 15 and 9 × 3 = 27, so **15 to 27**.
3. 50 − 9 = 41 and 50 − 5 = 45, so **41 to 45**.
4. 5 + 2 = 7 and 9 + 4 = 13, so **7 to 13**.
5. The note was wrong. Someone made a mistake about the clue, and it is time to check where.

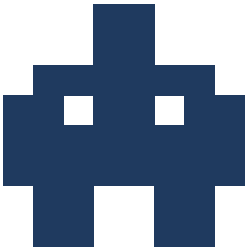
Spot the Slip.
Notebook A: slip on line **5**. 3 × 3 is 9. The answer should be 1,591.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 399.
Notebook C: **perfect**. 64 × 76 = 4,864.

Answers (turn the book around)

Chapter 5

1. $30 - 7 = \mathbf{23}$. Check: $23 + 7 = 30$. ✓
2. Undo +5: $29 - 5 = 24$. Undo ×3: $24 \div 3 = \mathbf{8}$. Check: $8 \times 3 + 5 = 29$. ✓
3. Undo ×2: $18 \div 2 = 9$. Undo −4: $9 + 4 = \mathbf{13}$.
4. $((13 - 4) \times 2) \rightarrow (9 \times 2) \rightarrow 18$. ✓
5. No. Many pairs fit, such as 5 and 15 or 10 and 10. A clue about x, or about y, would help.

Color by Answer (2,475). The shaded boxes make a robot.



Chapter 6

1. $((2 \times x) + (2 \times 5)) \rightarrow ((2 \times x) + 10)$.
2. No. At $x = 0$ the first gives 5 and the second gives 10.
3. Yes. Both mean two lots of x, so $(x + x)$ and $(2 \times x)$ are always equal.
4. $(4 \times (x + 3))$, which is Sam's note.
5. They agree at **0 and 2** only.

Balloon Message.

1.	$28 - 14$	$= 14$	N
2.	$4 - 3$	$= 1$	A
3.	$16 - 3$	$= 13$	M
4.	$26 - 21$	$= 5$	E
5.	$9 \times 2 + 2$	$= 20$	T
6.	$5 \times 2 - 2$	$= 8$	H
7.	$5 \times 2 - 5$	$= 5$	E
8.	$(7 - 4) \times 7$	$= 21$	U
9.	$(13 + 6) - 5$	$= 14$	N
10.	$(9 + 9) - 7$	$= 11$	K
11.	$(13 - 6) \times 2$	$= 14$	N
12.	$4 \times 4 - 1 \times 1$	$= 15$	O
13.	$12 \times 12 - 11 \times 11$	$= 23$	W
14.	$7 \times 5 - 7 \times 3$	$= 14$	N

Message: **NAME THE UNKNOWN!**

Answers (turn the book around)

Chapter 7

1. 11 ÷ 4 is 2 with 3 left over, so 2 + 3/4. The fraction pops to 0.75, so the answer is **2.75**.
2. 3 + 1/3. No: as a decimal it never ends, so keep **1/3**.
3. (3 × 3) + (3 × 1/3) = 9 + 1 = **10**. ✓
4. 7 goes into 22 three times with 1 left, so 3 + 1/7. Check: (7 × 3) + 1 = 22. ✓
5. When you only need a rough answer and say it is approximate. For an exact answer, keep the fraction.

Spot the Slip.
Notebook A: slip on line **5**. 2 × 2 is 4. The answer should be 1,596.
Notebook B: **perfect**. 24 × 26 = 624.
Notebook C: slip on line **3**. The corner is 4 × 4, not 4 × (4 + 1). The answer should be 2,484.

Chapter 8

1. d = 1 gives 2,499, d = 2 gives 2,496, and d = 3 gives 2,491. The range is **2,491 to 2,499**.
2. d = 2 gives **2,496**, and d = 4 gives **2,484**.
3. (a) **Has a reason**: the smallest product is 2,475. (b) **Only seen so far**: it is true when d is 3, but nothing says so. (c) **Refuted**: a corner is always taken away.
4. The note is (900 − (d × d)). d = 1 gives 899, and d = 2 gives 896. The range is **896 to 899**.
5. The note says exactly what is known and can be finished later. “About 2,490” is one guess inside that range, and it throws away the structure that says why the product is where it is.

Color by Answer (2,496). The shaded boxes make a robot.

